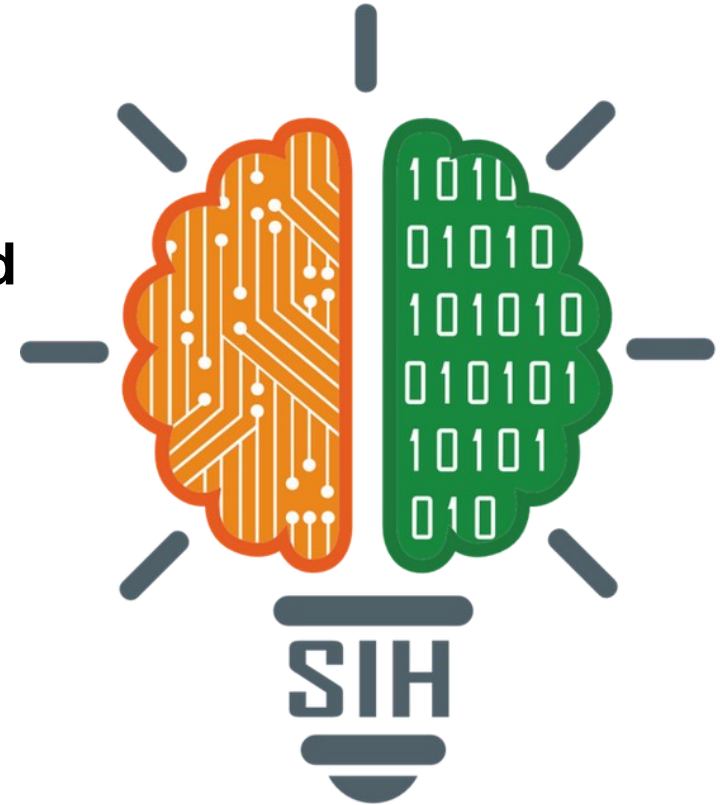


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26210**
- **Problem Statement Title- Student Innovation- Development solutions, keeping in mind the need to enhance the primary sector of India – Agriculture and to manage and process our agriculture produce.**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Hardware**
- **Team ID- 159400**
- **Team Name: INVICTUS**



NER-SmartCool: Solar-Powered Smart Mini Cold Storage

Proposed Solution

A **decentralized, solar-powered mini cold-storage unit** for farm-gate, village collection centres and FPOs in NER: a **temporary storage buffer** between harvest and transport.

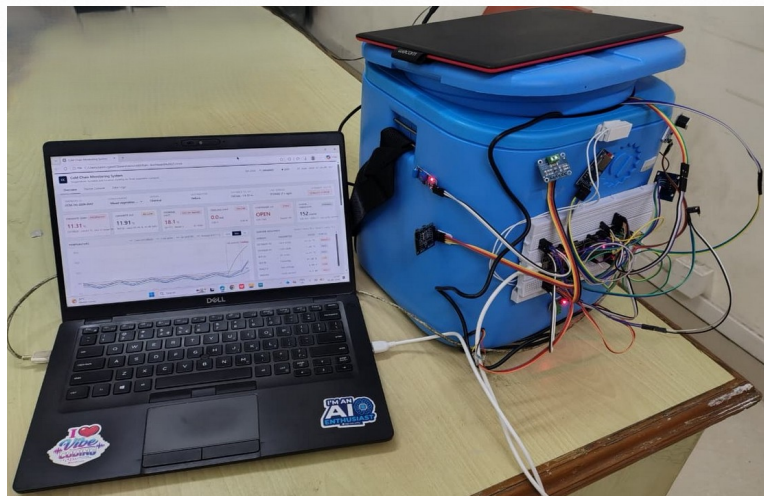
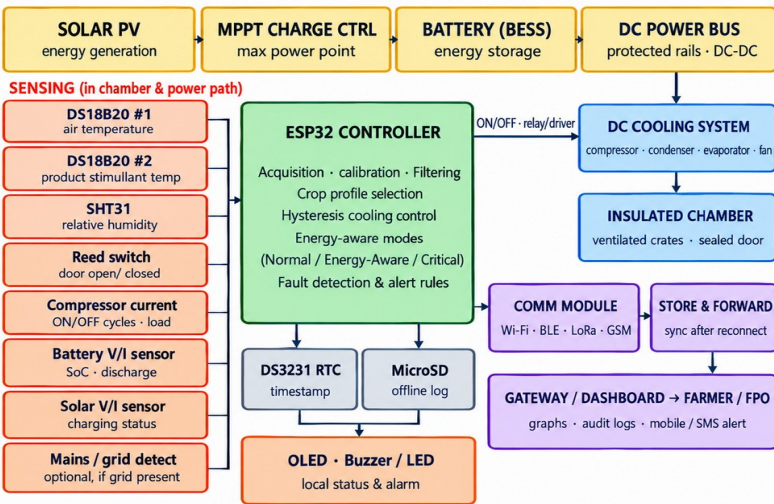
Insulated chamber + DC compressor refrigeration + forced-air circulation, fed by **Solar PV** → **MPPT** → **Battery** → **DC bus** for unreliable-grid and off-grid sites.

ESP32 closed-loop control with hysteresis holds crop-specific temperature and humidity; **energy-aware modes** (Normal / Energy-Aware / Critical) protect produce on low battery.

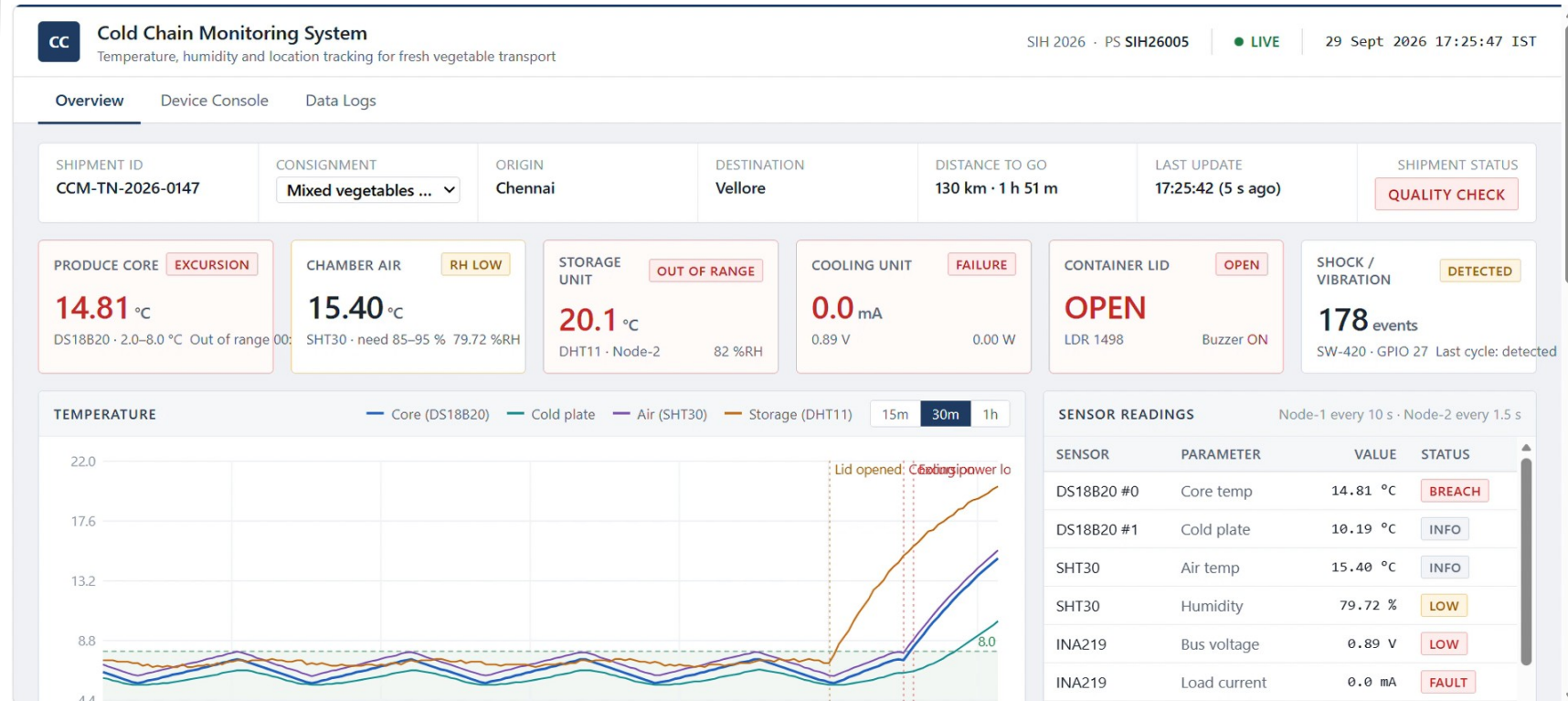
Fault detection for high temperature, door-open, low battery, solar-charging and cooling failure, with **local buzzer / LED / display alerts** that need no internet.

Offline-first: RTC + MicroSD logging with **store-and-forward sync** (Wi-Fi and GSM) and a simple crop-select interface. Exact set-points to be validated.





Prototype



Dashboard

Techstack Used:

1. Solar PV + MPPT controller
2. Battery (BESS) + DC protection
3. DC compressor refrigeration
4. PU/PIR insulated chamber
5. ESP32 (Arduino / ESP-IDF C++)
6. Digital temp + RH sensors
7. Reed switch, V/I sensors
8. RTC + MicroSD, OLED/LCD
9. Wi-Fi / GSM / LoRa (optional)
10. Web dashboard (optional)

Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
Technical – Achievable	Sizing solar, battery and cooling load for NER climate	Cooling-load and energy calculations before hardware selection; proven DC refrigeration, MPPT and standard sensors ; ratings validated on the prototype.
Reliability – Resilient	Cloudy / rainy days, night, grid outages, no network	Solar + battery with energy-aware / critical modes ; offline-first control, alerts and logging; fault detection for cooling, door and solar failure.
Operational – Practical	Unreliable power & connectivity; limited cold-storage access; limited technical support	Local data storage + cloud synchronization, Safe Holdover Time Prediction , intelligent alerts, farmer-friendly crop selection, local status display, offline-first operation, weather-resistant modular unit .

Given that feasibility and viability are our top priority we will ensure to keep everything

- **Technically viable** – uses existing solar, DC refrigeration and ESP32 technology; one or two crops validated first.
- **Economically feasible** – small modular units for FPO / collection-centre deployment; cost to be worked out after design.
- **Environmentally sound** – renewable solar energy, energy-efficient cooling, designed to cut post-harvest and transport losses.

Potential Impact on Target Audience

1. Farmers

- Designed to reduce forced immediate sale and spoilage; flexible transport timing.

2. FPOs & Cooperatives

- Shared, monitored local storage at aggregation points.

3. Village collection centres / traders

- Better produce handling before transport to markets.

4. Remote NER communities

- Cold-chain access without depending on grid supply.

5. Government / MDoNER

- Scalable decentralized horticulture cold-chain model.

Benefits of the solution

1. Social

- Improves farmer bargaining flexibility; supports cooperatives.

2. Economic

- Expected to improve income opportunity by cutting losses.

3. Environmental

- Renewable solar energy; less wasted produce and transport.

4. Technological

- Energy-aware, offline-first rural embedded system.

5. Strategic

- Replicable modular units across horticultural clusters.

All impacts are design intents, to be validated through prototype testing:

Temperature stability

Cooling response

Solar charging

Battery backup

Safe Holdover Time
Prediction

Alert response

Offline log retention

Vegetable quality check

Research Survey

Aspect	Conventional Systems	Proposed System
Location	Large centralized cold rooms, far from farms	Decentralized farm-gate / village mini units
Power	Grid-dependent	Solar + battery, energy-aware operation
Control	Basic thermostat	Closed-loop, crop-specific profiles
Monitoring	Manual checks, network-dependent	Offline-first logging + local alerts
Faults	Found after damage	Cooling / door / solar fault detection

Reference

- [1] FAO, “Global Food Losses and Food Waste – Extent, Causes and Prevention”, Rome (2011).
- [2] ICAR-CIPHET, “Assessment of Quantitative Harvest and Post-Harvest Losses of Major Crops and Commodities in India”, Ludhiana (2015).
- [3] National Centre for Cold-chain Development (NCCD) and National Horticulture Board (NHB): cold-chain infrastructure and horticulture data.
- [4] MNRE / NABARD: solar-powered cold storage and agri-infrastructure guidelines.
- [5] Peer-reviewed literature on solar PV–battery DC refrigeration and small-scale vegetable cold storage (to be shortlisted).

Note: sources are institutional; exact editions and links to be verified before submission. No performance figures are claimed.